Chapter 5

redox titration

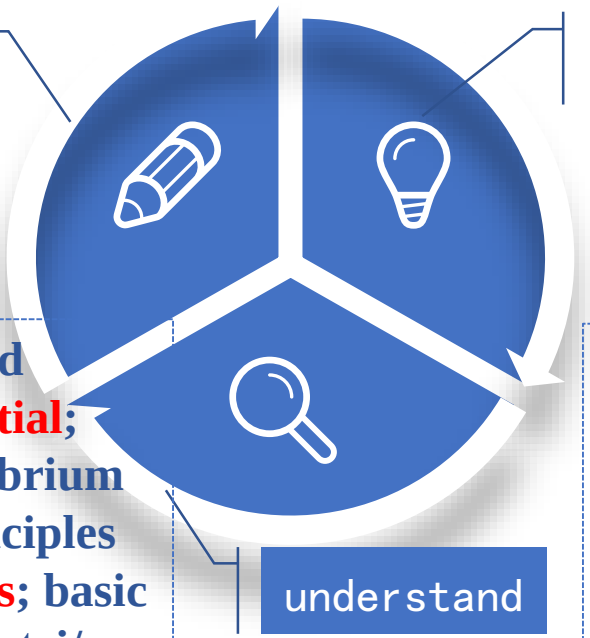
Tongbo Wu

Department of Pharmaceutical Analysis

Contents

- Section 1 Overview
- Section 2 Basic Principles of Redox Titration
- Section 3 Common Redox Titration Methods

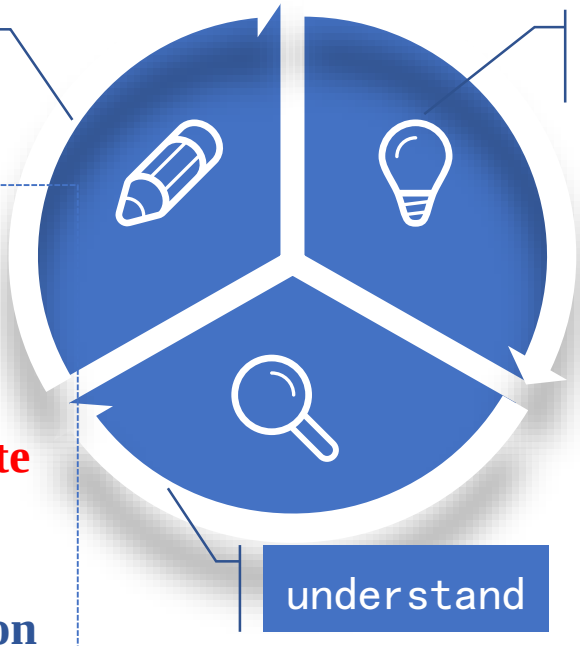
Learning Objectives



concepts, influencing factors and calculation of **conditional potential**; calculation of conditional equilibrium constant of **redox reaction**; principles and selection of redox **indicators**; basic principles of iodimetry/ˌaɪəˈdɪmɪtri/, permanganometry/ˌpɜːmæŋgəˈnɒmɪtri/ and sodium nitrite titration; calculations in **redox titration**

- **条件电位**的概念、影响因素和计算；**氧化还原**反应条件平衡常数的计算；氧化还原**指示剂**指示终点的原理和选择原则；碘量法、高锰酸钾法和亚硝酸钠法的基本原理；**氧化还原滴定**分析的计算

Learning Objectives



Mast

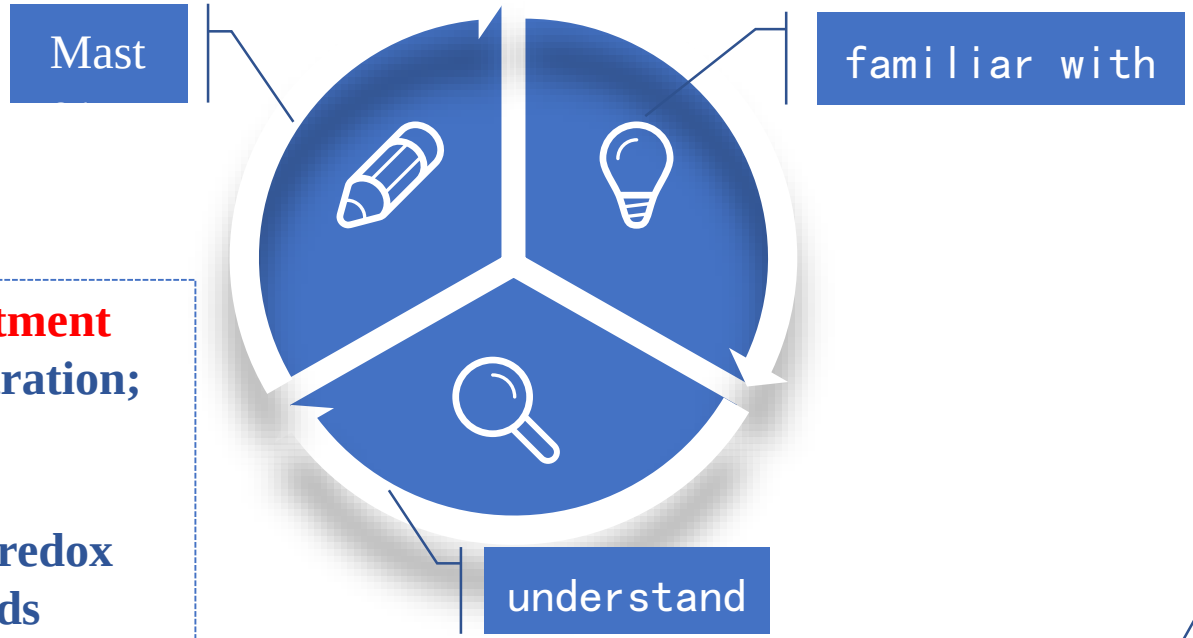
redox titration curves; factors affecting **potential jump** and calculation of **jump range**; factors affecting reaction rate; principles of **potassium bromate** /ˈbrəʊmeɪt/ and **bromine** /ˈbrəʊmiːn/ methods; preparation and standardisation of iodine and sodium thiosulfate solutions

familiar with 熟悉

- 氧化还原滴定曲线、影响**电位突跃**范围的因素和突跃范围的计算；影响氧化还原反应速度的因素；**溴酸钾法和溴量法**等的基本原理；碘和硫代硫酸钠标准溶液的配制和标定

understand

Learning Objectives



- sample **pretreatment** before redox titration; **endpoint error** calculation; applications of redox titration methods

- 氧化还原滴定前的试样**预处理**；滴定**终点误差**计算；各种氧化还原滴定法的应用

Section 1 Overview

Section 1 Overview

1 Redox titration

redox titration (redox titration）is based on redox reaction titrimetric analysis method; the reaction mechanism and process are complex, somethe reaction rate is slow and side reactions副反应 often occur; the reaction medium介质 greatly affects the process

2 Common redox titration methods

iodimetry, permanganometry, sodium nitrite titration, dichromate titration, bromate titration, cerimetry, etc.

碘量法、高锰酸钾法、亚硝酸钠法、重铬酸钾法、溴酸钾法、铈量法等

Section 2

basic principles of redox titration

Section 2 Basic Principles of Redox Titration

1 Redox equilibrium氧化还原平衡

(1) Redox couple氧化还原电对

Reversible couple: can rapidly establish redox equilibrium, $\varphi_{measured} = \varphi_{theory}$

Fe^{3+}/ Fe^{2+}、I_2/I^-

Irreversible couple: cannot rapidly establish redox equilibrium, $\varphi_{measured} \neq \varphi_{theory}$

MnO_4^-/ Mn^{2+}、$Cr_2O_7^{2-}/ Cr^{3+}$、$S_4O_6^{2-}/S_2O_3^{2-}$

可逆电对：能迅速建立氧化还原平衡，$\varphi_{测得} = \varphi_{理论}$

不可逆电对：不能迅速建立氧化还原平衡，$\varphi_{测得} \neq \varphi_{理论}$

Section 2 Basic Principles of Redox Titration

1 Redox equilibrium

(1) Redox couple 对称电对：氧化态与还原态的系数相同

Symmetric couple: the coefficients of the oxidised and reduced forms are equal

$Fe^{3+} + e \rightleftharpoons Fe^{2+}$ ， $MnO_4^- + 8H^+ + 5e \rightleftharpoons Mn^{2+} + 4H_2O$, that is Fe^{3+}/Fe^{2+} couple andMnO_4^-/Mn^{2+}couples are symmetric

Asymmetric couple: the coefficients of the oxidised and reduced forms are different

$I_2 + 2e \rightleftharpoons 2I$， $Cr_2O_7^{2-} + 6I + 14H^+ \rightleftharpoons 2Cr^{3+} + 3I_2 + 7H_2O$ etc., that isI_2/I^-couple and$Cr_2O_7^{2-}/Cr^{3+}$couples are asymmetric

不对称电对：氧化态与还原态的系数不同

Section 2 Basic Principles of Redox Titration

1 Redox equilibrium

可逆电对的电极电位可用Nernst方程式表示，例如，以Ox表示氧化态，Red表示还原态，可逆氧化还原电对Ox/ Red的半电池反应为：

(1) Redox couple

the electrode potential of a reversible couple can be expressed by Nernst equation, for example, using Ox represents the oxidised form, Red represents the reduced form; reversible redox coupleOx/ Red half-cell reaction is:

Nernst equation:

$$a\mathrm{Ox} + ne \rightleftharpoons b\mathrm{Red}$$

$$\varphi_{\mathrm{Ox/Red}} = \varphi^{\ominus} + \frac{2.303RT}{nF}\lg\frac{a_{\mathrm{Ox}}^{a}}{a_{\mathrm{Red}}^{b}}$$

$$= \varphi^{\ominus} + \frac{0.059}{n}\lg\frac{a_{\mathrm{Ox}}^{a}}{a_{\mathrm{Red}}^{b}} \quad (25\,℃)$$

$\varphi^{\ominus}$is the standard electrode potential标准电极电位, a_{Ox} is the activity氧化态活度of the oxidised form, $a_{\mathrm{Red}}$is the activity of the reduced form还原态活度

Section 2 Basic Principles of Redox Titration

1 Redox equilibrium

浓度c，活度系数γ，副反应系数α

(2) Conditional potential

条件电位

$$\varphi_{Ox/Red} = \varphi^{\ominus} + \frac{0.059}{n}\lg\frac{\gamma_{Ox}c_{Ox}}{\alpha_{Ox}}\cdot\frac{\alpha_{Red}}{\gamma_{Red}c_{Red}}$$

$$= \left(\varphi^{\ominus} + \frac{0.059}{n}\lg\frac{\gamma_{Ox}\alpha_{Red}}{\gamma_{Red}\alpha_{Ox}}\right) + \frac{0.059}{n}\lg\frac{c_{Ox}}{c_{Red}}$$

$$= \varphi^{\ominus'} + \frac{0.059}{n}\lg\frac{c_{Ox}}{c_{Red}}$$

$$\varphi^{\ominus'} = \varphi^{\ominus} + \frac{0.059}{n}\lg\frac{\gamma_{Ox}\alpha_{Red}}{\gamma_{Red}\alpha_{Ox}}$$

这里把上一页指数里的a和b都视为1了

电对的氧化态和还原态的分析浓度均为1mol/L时或它们的分析浓度的比值为1时的实际电位

The actual potential when the analytical concentrations of the oxidized and reduced forms of the redox couple are both 1 mol/L, or when the ratio of their analytical concentrations equals 1

Section 2 Basic Principles of Redox Titration

1 Redox equilibrium

(3) Factors affecting the conditional potential

1. salt effect盐效应

$change\ in\ ionic\ strength \Rightarrow ionic\ activity\ coefficient\ (\gamma)\ change \Rightarrow \varphi^{\ominus\prime} changes$

$$\varphi_{\ominus}' = \varphi^{\ominus} + \frac{0.059}{n} \lg \frac{\gamma_{Ox}}{\gamma_{Red}} \qquad (25℃)$$

$the\ salt\ effect\ is\ difficult\ to\ calculate\ and\ can\ be\ neglected \Rightarrow assume \gamma = 1$

Section 2 Basic Principles of Redox Titration

1 Redox equilibrium

$$\varphi^{\ominus'} = \varphi^{\ominus} + \frac{0.059}{n}\lg\frac{\gamma_{Ox}\alpha_{Red}}{\gamma_{Red}\alpha_{Ox}}$$

(3) Factors affecting the conditional potential

2. precipitation生成沉淀

$oxidationformprecipitation \Rightarrow \varphi^{\ominus'}\downarrow$ 氧化态生成沉淀

$reductionformprecipitation \Rightarrow \varphi^{\ominus'}\uparrow$ 还原态生成沉淀

Example: determination of Cu^{2+} content by indirect iodimetry间接碘量法 is based on the following reaction:

$Cu^{2+} + e \rightleftharpoons Cu^{+}$ $\varphi^{\ominus}_{Cu^{2+}/Cu^{+}}=0.16V$

$I_2 + 2e \rightleftharpoons 2I^-$ $\varphi^{\ominus}_{I_2/I^-}=0.54V$

Theoretically

理论上 $I_2 + 2Cu^{+} \rightleftharpoons 2Cu^{2+} + 2I^-$

实际上 $2Cu^{2+} + 4I^- \rightleftharpoons 2CuI\downarrow + I_2$

actually

$Cu^{+} + I^- \rightarrow CuI$

α_{Red}变大

$\varphi^{\ominus}_{Cu^{2+}/Cu^{+}}\uparrow$

Section 2 Basic Principles of Redox Titration

Example 5-1: Calculate the conditional electrode potential of the Cu^{2+}/Cu^{+} couple at 25 ℃ when $[I^-] = 1$ mol/L. 忽略离子强度的影响

$$K_{sp}=[Cu^+][I^-]=1.1\times10^{-12},\quad \varphi^{\ominus}_{I_2/I^-}=0.54V,\quad \varphi^{\ominus}_{Cu^{2+}/Cu^+}=0.16V$$

$$\varphi_{Cu^{2+}/Cu^+}=\varphi^{\ominus}+0.059\lg\frac{[Cu^{2+}]}{[Cu^+]}\qquad \varphi_{Cu^{2+}/Cu^+}=\varphi^{\ominus}+0.059\lg\frac{[Cu^{2+}][I^-]}{K_{sp}}$$

若考虑到副反应影响，则有：$[Cu^{2+}]=c_{Cu^{2+}}/\alpha_{Cu^{2+}}$

$$\varphi_{Cu^{2+}/Cu^+}=\varphi^{\ominus}+0.059\lg\frac{[I^-]}{K_{sp}\alpha_{Cu^{2+}}}+0.059\lg c_{Cu^{2+}}=\varphi^{\ominus\prime}+0.059\lg c_{Cu^{2+}}$$

$$\varphi^{\ominus}_{Cu^{2+}/Cu^+}=\varphi^{\ominus}+0.059\lg\frac{[I^-]}{K_{sp}\alpha_{Cu^{2+}}}$$

因为在实验条件下 Cu^{2+}不发生明显的副反应，$\alpha_{Cu^{2+}}\approx1$，$[I^-]=1$mol/L，则

$$\varphi^{\ominus}_{Cu^{2+}/Cu^+}=0.16+0.059\lg\frac{1}{1.1\times10^{-12}\times1}=0.87\ (V)$$

Section 2 Basic Principles of Redox Titration

1 Redox equilibrium

$$\varphi^{\ominus'} = \varphi^{\ominus} + \frac{0.059}{n} \lg \frac{\gamma_{Ox}\alpha_{Red}}{\gamma_{Red}\alpha_{Ox}}$$

(3) Factors affecting the conditional potential

3.complex formation生成配合物

forms complexes with both

- stability of the oxidised complex > stability of the reduced complex $\rightarrow \varphi^{\ominus'}\downarrow$
- stability of the reduced complex > stability of the oxidised complex $\rightarrow \varphi^{\ominus'}\uparrow$

与两者均形成配合物

- 氧化态配合物的稳定性>还原态配合物的稳定性$\rightarrow \varphi^{\ominus'}\downarrow$
- 还原态配合物的稳定性>氧化态配合物的稳定性$\rightarrow \varphi^{\ominus'}\uparrow$

Section 2 Basic Principles of Redox Titration

Example: Calculate the conditional electrode potential of the Fe^{3+}/Fe^{2+} couple in a solution with pH = 3.0 and NaF concentration of 0.10 mol/L at 25℃.

$\varphi^{\ominus}_{Fe^{3+}/Fe^{2+}}$ =0.771V， FeF_3 $\lg\beta_1$, $\lg\beta_2$ and $\lg\beta_3$ are 5.2, 9.2 and 11.9, HF's K_a= 6.3×10^{-4}

$$\text{pH=3.0，}\ [F^-] = \delta_{F^-} \cdot c_{NaF} = \frac{K_a \cdot c_{NaF}}{[H^+]+K_a} = \frac{6.3\times10^{-4}\times0.10}{10^{-3.0}+6.3\times10^{-4}} = 10^{-1.41}$$

$$\alpha_{Fe^{3+}(F^-)} = 1 + \beta_1[F^-] + \beta_2[F^-]^2 + \beta_3[F^-]^3$$

$$= 1 + 10^{5.2} \times 10^{-1.41} + 10^{9.2} \times (10^{-1.41})^2 + 10^{11.9} \times (10^{-1.41})^3$$

$$\approx 10^{7.69}$$

Section 2 Basic Principles of Redox Titration

Fe^{2+} does not form stable complexes with F^-, $\alpha_{Fe^{2+}(F^-)} = 1$, conditional potential of the couple $Fe^{3+}/\ Fe^{2+}$:

$$\varphi^{\ominus}_{Fe^{3+}/Fe^{2+}} = \varphi^{\ominus} + 0.059\lg\frac{\alpha_{Fe^{2+}}}{\alpha_{Fe^{3+}}}$$

$$= 0.771 + 0.0591\lg\frac{1}{10^{7.69}}$$

$$= 0.317V$$

Section 2 Basic Principles of Redox Titration

1 Redox equilibrium

4. acid effect酸效应

If H^+ or OH^- participates in the half-cell reaction of an electrode couple, changes in the acidity of the solution will directly alter the conditional potential. When the oxidized or reduced forms of some substances are weak acids or weak bases, the solution acidity will also affect their existing forms, thereby causing variations in the conditional potential.

电对的半电池反应中若有H^+或 OH参加，溶液酸度改变将直接引起条件电位的改变；一些物质的氧化态或还原态若是弱酸或弱碱，溶液酸度还会影响其存在的形式，从而引起条件电位的变化

Section 2 Basic Principles of Redox Titration

1 Redox equilibrium

Example5-3 Calculate the conditional potential of the redox couple $H_3AsO_4/HAsO_2$ at 25℃ when $[H^+]$ = 5 mol/L or pH = 8.0, and determine the direction of the following reaction under the above two conditions.

$$H_3AsO_4 + 2H^+ + 2e \rightleftharpoons HAsO_2 + 2H_2O \qquad \varphi^{\ominus}_{H_3AsO_4/HAsO_2} = 0.56V$$

$$I_2 + 2e \rightleftharpoons 2I^- \qquad \varphi^{\ominus\prime}_{I_2/I^-} \approx \varphi^{\ominus}_{I_2/I^-} = 0.54V$$

$$\varphi_{H_3AsO_4/HAsO_2} = \varphi^{\ominus} + \frac{0.059}{2}\lg\frac{[H_3AsO_4][H^+]^2}{[HAsO_2]} = \varphi^{\ominus} + \frac{0.059}{2}\lg\frac{c_{H_3AsO_4}\alpha_{HAsO_2}[H^+]^2}{c_{HAsO_2}\alpha_{H_3AsO_4}}$$

Section 2 Basic Principles of Redox Titration

1 Redox equilibrium

4.acid effect

$$\varphi_{H_3AsO_4/HAsO_2} = \varphi^{\ominus'} + \frac{0.059}{2}\lg\frac{c_{H_3AsO_4}}{c_{HAsO_2}} \qquad \varphi^{\ominus'} = \varphi^{\ominus} + \frac{0.059}{2}\lg\frac{\alpha_{HAsO_2}[H^+]^2}{\alpha_{H_3AsO_4}}$$

$$\alpha_{H_3AsO_4} = \frac{[H^+]^3 + [H^+]^2 \cdot K_{a_1} + [H^+] \cdot K_{a1}K_{a2} + K_{a1}K_{a2}K_{a3}}{[H^+]^3}$$

$$\alpha_{HAsO_2} = \frac{[H^+]+K_a}{[H^+]}$$

$HAsO_2$ K_a 5.1×10^{-10}, H_3AsO_4 $K_{a1}\sim pK_{a3}$ *are* 5.5×10^{-3}、1.7×10^{-7}、5.1×10^{-12}

Section 2 Basic Principles of Redox Titration

1 Redox equilibrium

氧化剂Ox1+还原剂Red2 <-> 还原产物Red1+氧化产物Ox2
$\varphi^{\ominus\prime}$大的氧化还原对里的氧化态做氧化剂
$\varphi_1^{\ominus\prime}$大正向反应，$\varphi_2^{\ominus\prime}$大逆向反应

4.acid effect

总反应 $HAsO_2 + I_2 + 2H_2O \rightleftharpoons H_3AsO_4 + 2H^+ + 2I^-$

当$[H^+] = 5mol/L$时，$\alpha_{HAsO_2} = 1.0$，$\alpha_{H_3AsO_4} = 1.0$，$\varphi^{\ominus\prime}_{H_3AsO_4/HAsO_2} = 0.60V$，由于$\varphi^{\ominus\prime}_{H_3AsO_4/HAsO_2} > \varphi^{\ominus\prime}_{I_2/I^-}$，故反应向左进行。因此，可用间接碘量法在强酸性溶液中测定H_3AsO_4。

当$pH = 8.0$时，$\alpha_{HAsO_2} = 1.0$，$\alpha_{H_3AsO_4} = 1.0 \times 10^7$，$\varphi^{\ominus\prime}_{H_3AsO_4/HAsO_2} = -0.12V$，则$\varphi^{\ominus\prime}_{H_3AsO_4/HAsO_2} < \varphi^{\ominus\prime}_{I_2/I^-}$，故反应向右进行。因此，可用$As_2O_3$在弱碱性溶液中标定$I_2$标准溶液。

根据电对的电位高低判断氧化还原反应的方向。

judge the direction of a redox reaction from electrode potentials

Section 2 Basic Principles of Redox Titration

1 Redox equilibrium

(4) Completeness of the redox reaction氧化还原反应的完全程度

1.conditional equilibrium constant条件平衡常数K'

$$Ox_1 + n_1e \rightleftharpoons Red_1 \qquad \varphi_1=\varphi_1^{\ominus'}+\frac{0.059}{n_1}\lg\frac{c_{Ox_1}}{c_{Red_1}}$$

$$Ox_2 + n_2e \rightleftharpoons Red_2 \qquad \varphi_2=\varphi_2^{\ominus'}+\frac{0.059}{n_2}\lg\frac{c_{Ox_2}}{c_{Red_2}}$$

at equilibrium, $\varphi_1=\varphi_2$, that is

$$\varphi_1^{\ominus'}+\frac{0.059}{n_1}\lg\frac{c_{Ox_1}}{c_{Red_1}}=\varphi_2^{\ominus'}+\frac{0.059}{n_2}\lg\frac{c_{Ox_2}}{c_{Red_2}}$$

Section 2 Basic Principles of Redox Titration

1 Redox equilibrium

(4) Completeness of the redox reaction

1.conditional equilibrium constant K' 设n是两电对电子转移数n_1与n_2的最小公倍数

Let n is the number of electrons transferred by the two couples $n_1$and n_2 least common multiple, then $n_1=n/a$， $n_2=n/b$, substituting gives:

$$\varphi_1=\varphi_1^{\ominus\prime}+\frac{0.059}{n_1}\lg\frac{c_{Ox_1}}{c_{Red_1}}$$

$$\varphi_2=\varphi_2^{\ominus\prime}+\frac{0.059}{n_2}\lg\frac{c_{Ox_2}}{c_{Red_2}}$$

$$\varphi_1^{\ominus\prime}+\frac{0.059}{n_1}\lg\frac{c_{Ox_1}}{c_{Red_1}}=\varphi_2^{\ominus\prime}+\frac{0.059}{n_2}\lg\frac{c_{Ox_2}}{c_{Red_2}}$$

$$\frac{0.059}{n}\lg\frac{c_{Ox_2}^{b}c_{Red_1}^{a}}{c_{Ox_1}^{a}c_{Red_2}^{b}}=\varphi_1^{\ominus\prime}-\varphi_2^{\ominus\prime}$$

$$\lg K'=\lg\frac{c_{Ox_2}^{b}c_{Red_1}^{a}}{c_{Ox_1}^{a}c_{Red_2}^{b}}=\frac{n(\varphi_1^{\ominus\prime}-\varphi_2^{\ominus\prime})}{0.059}=\frac{n\Delta\varphi^{\ominus\prime}}{0.059}$$

Section 2 Basic Principles of Redox Titration

1 Redox equilibrium

(4) Completeness of the redox reaction

1.conditional equilibrium constant K'

$$\lg K' = \frac{n\Delta\varphi^{\ominus'}}{0.059}$$

The larger the conditional potential difference $\Delta\varphi^{\ominus'}$ between the two redox couples and the greater the number of electrons n gained or lost in the reaction, the larger the value of K′ (or lg K′), and the more completely the reaction proceeds to the right.

Section 2 Basic Principles of Redox Titration

1 Redox equilibrium

(4) Completeness of the redox reaction

2.extent of reaction

according to titration error requirements, reaction completeness> 99.9%

$$\lg K' = \lg \frac{c_{Ox_2}^{b} c_{Red_1}^{a}}{c_{Ox_1}^{a} c_{Red_2}^{b}} = \lg \frac{(99.9\%)^{b} (99.9\%)^{a}}{(0.1\%)^{a} (0.1\%)^{b}}$$

$$\approx \lg(10^{3a} \times 10^{3b}) = 3(a+b)$$

$$\triangle \varphi^{\ominus'} = \frac{0.059}{n} \lg K' = \frac{0.059 \times 3(a+b)}{n}$$

$\lg K' \geqslant 3(a+b)$ 或 $\triangle \varphi^{\ominus'} \geqslant 0.059 \times 3(a+b)/n$

Section 2 Basic Principles of Redox Titration

1 Redox equilibrium

$\lg K' \geqslant 3(a+b)$ 或 $\Delta\ \varphi^{\ominus'} \geqslant 0.059 \times 3\ (a+b)/n$

(4) Completeness of the redox reaction

2.extent of reaction

（1）1∶1 type redox reaction, $n_1 = n_2 = 1, \lg K' \geqslant 6\ or\ \Delta\varphi' \geqslant 0.35V$;

$$n_1 = n_2 = 2, \lg K' \geqslant 6\ or\ \Delta\varphi' \geqslant 0.18V$$

（2）1∶2 type redox reaction, $n_1 = 2$， $n_2 = 1, \lg K' \geqslant 9\ or\ \Delta\varphi' \geqslant 0.27V$

Section 2 Basic Principles of Redox Titration

Example: Determine whether the reaction can proceed completely when titrating Fe^{2+} solution with Ce^{4+} solution in 1 mol/L H_2SO_4 solution.（$\varphi^{\ominus'}_{Ce^{4+}/Ce^{3+}}$=1.44V，$\varphi^{\ominus'}_{Fe^{3+}/Fe^{2+}}$=0.68V）

the titration reaction $Ce^{4+} + Fe^{2+} \rightleftharpoons Ce^{3+} + Fe^{3+}$ belongs to 1：1 type redox reaction ($n_1 = n_2 = 1$）

$$\triangle \varphi^{\ominus'} = 1.44 - 0.68 = 0.76V > 0.4V$$

$$\lg K' = \frac{n\Delta\varphi^{\ominus'}}{0.059} = \frac{1\times 0.76}{0.059} = 12.9$$

$$K' = 7.6 \times 10^{12}$$

$$\lg K' \geqslant 3(a+b) \text{ 或 } \triangle \varphi^{\ominus'} \geqslant 0.059\times 3(a+b)/n$$

Considering only the conditional equilibrium constant, the above reaction can proceed completely and can be applied to redox titration analysis.

Section 2 Basic Principles of Redox Titration

1 Redox equilibrium

(5) Rate of redox reaction

1. Effect of concentration on reaction rate

increasing reactant concentration accelerates reaction

Example $Cr_2O_7^{2-} + 6I^- + 14H^+ \rightleftharpoons 2Cr^{3+} + 3I_2 + 7H_2O$

$Note$: $[H^+] \uparrow$, $[I^-] \uparrow$, $v \uparrow$

Section 2 Basic Principles of Redox Titration

1 Redox equilibrium

(5) Rate of redox reaction

2. Temperature effect on reaction rate

generally, increasing temperature increases reaction rate; Usually, for every 10℃ rise in temperature, the reaction rate increases by about 2 to 3 times.

Example $2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightleftharpoons 2Mn^{2+} + 10CO_2 + 8H_2O$

Very slow at room temperature, reaction temperature 75～85℃

Substances that are volatile and susceptible to oxidation are not applicable.

易挥发和易氧化物质不适用

Section 2 Basic Principles of Redox Titration

1 Redox equilibrium

(5) Rate of redox reaction

3.effect of catalyst催化剂 on reaction rate

Catalysts can fundamentally alter the reaction pathway and reaction rate.

Catalysts that speed up the reaction rate are called positive catalysts (catalyst);

Catalysts that slow down the reaction rate are called negative catalysts (anticatalyst).

Section 2 Basic Principles of Redox Titration

2 Redox titration curve氧化还原滴定曲线

Definition

A redox titration curve is plotted with the volume of titrant added (or titration percentage) as the abscissa and the electrode potential of the relevant redox couple as the ordinate.

以滴定剂加入体积（或滴定百分数）为横坐标，以相关电对的电极电位为纵坐标作图，所得曲线即氧化还原滴定曲线。

Section 2 Basic Principles of Redox Titration

2 Redox titration curve氧化还原滴定曲线

Why plot it?

During titration, the concentrations of the oxidizing agent and reducing agent keep changing, and the potentials of the two redox pairs change accordingly. The jump range of the curve serves as the basis for selecting an indicator.

滴定进行中，氧化剂与还原剂浓度不断改变，两电对电位随之改变。曲线的突跃范围正是选择指示剂的依据。

Section 2 Basic Principles of Redox Titration

2 Redox titration curve氧化还原滴定曲线

System conditions体系条件

- Medium 介质：1 mol/L H_2SO_4
- Titrant 滴定剂：0.1000 mol/L $Ce(SO_4)_2$
- Test solution 被测液：20.00 mL 0.1000 mol/L $FeSO_4$
- Conditional equilibrium constant 条件平衡常数 $K' = 7.6 \times 10^{12}$， the reaction can proceed completely 反应可完全进行

Reaction: $Ce^{4+} + Fe^{2+} \rightleftharpoons Ce^{3+} + Fe^{3+}$

Section 2 Basic Principles of Redox Titration

2 Redox titration curve氧化还原滴定曲线

Reaction: $Ce^{4+} + Fe^{2+} \rightleftharpoons Ce^{3+} + Fe^{3+}$

Titrant redox couple滴定剂电对

$Ce^{4+} + e^{-} \rightleftharpoons Ce^{3+}$

$\varphi^{\circ\prime} = 1.44\ \mathrm{V}$

$$\varphi = \varphi^{\ominus\prime}_{Ce^{4+}/Ce^{3+}} + 0.059\lg\frac{c_{Ce^{4+}}}{c_{Ce^{3+}}}$$

Test substance redox couple被测物电对

$Fe^{3+} + e^{-} \rightleftharpoons Fe^{2+}$

$\phi^{\circ\prime} = 0.68\ \mathrm{V}$

$$\varphi = \varphi^{\ominus\prime}_{Fe^{3+}/Fe^{2+}} + 0.059\lg\frac{c_{Fe^{3+}}}{c_{Fe^{2+}}}$$

Section 2 Basic Principles of Redox Titration

2 Redox titration curve氧化还原滴定曲线

① Before titration starts: a tiny amount of Fe^{3+} is produced as Fe^{2+} is oxidized by air, and $c(Fe^{3+})$ is unknown. The potential cannot be calculated

② Before the equivalence point: Nearly all Ce^{4+} is reduced to Ce^{3+}

The Fe^{3+}/Fe^{2+} redox couple is selected. The ratio $c(Fe^{3+})/c(Fe^{2+})$ can be obtained from the titration percentage. 19.98ml Ce^{4+} ,99.9% Fe^{3+} turns to Fe^{2+}

$$\varphi = \varphi^{\ominus\prime}_{Fe^{3+}/Fe^{2+}} + 0.059\lg\frac{99.9}{0.1}$$

$$\varphi = 0.68 + 0.059 \times 3 = 0.86\mathrm{V}$$

Reaction: $Ce^{4+} + Fe^{2+} \rightleftharpoons Ce^{3+} + Fe^{3+}$

为什么选 Fe 电对

此时加入的 Ce^{4+} 几乎全部还原为 Ce^{3+}，未反应 Ce^{4+} 浓度极小且难直接求得。

而由滴定百分数可直接确定 $c(Fe^{3+})/c(Fe^{2+})$，Nernst 式最易算。

Section 2 Basic Principles of Redox Titration

2 Redox titration curve氧化还原滴定曲线

Reaction: $Ce^{4+} + Fe^{2+} \rightleftharpoons Ce^{3+} + Fe^{3+}$

③ At the equivalence point: both Ce^{4+} and Fe^{2+} are quantitatively converted.

Neither half-reaction can be used alone. Add the two equations to solve for φ_{sp}.

$$c_{Ce^{4+}} = c_{Fe^{2+}}, \quad c_{Ce^{3+}} = c_{Fe^{3+}}$$

Nernst equation:

$$\varphi_{sp} = \varphi^{\ominus\prime}{}_{Ce} + 0.059\lg\frac{c_{Ce^{4+}}}{c_{Ce^{3+}}} \quad \cdots(1)$$

$$\varphi_{sp} = \varphi^{\ominus\prime}{}_{Fe} + 0.059\lg\frac{c_{Fe^{3+}}}{c_{Fe^{2+}}} \quad \cdots(2)$$

$$2\varphi_{sp} = \varphi^{\ominus\prime}{}_{Ce} + \varphi^{\ominus\prime}{}_{Fe} = 1.44 + 0.68 = 2.12\text{V}$$

$$\varphi_{sp} = \frac{1.44 + 0.68}{2} = 1.06\text{V}$$

Section 2 Basic Principles of Redox Titration

2 Redox titration curve氧化还原滴定曲线

Reaction: $Ce^{4+} + Fe^{2+} \rightleftharpoons Ce^{3+} + Fe^{3+}$

④ After the equivalence point: Fe^{2+} is almost completely oxidized.

Select the Ce^{4+}/Ce^{3+} redox couple.

Calculate the concentration ratio from the percentage of excess Ce^{4+}

Ce^{4+} 20.02 mL, excess 0.1%:

$$\varphi = 1.44 + 0.059\lg\frac{0.1}{100}$$

$$\varphi = 1.44 - 0.059 \times 3 = 1.26\text{V}$$

Section 2 Basic Principles of Redox Titration

2 Redox titration curve氧化还原滴定曲线

Reaction: $Ce^{4+} + Fe^{2+} \rightleftharpoons Ce^{3+} + Fe^{3+}$

滴入/mL	滴定/%	电位/V
5.00	25.0	0.60
10.00	50.0	0.65
19.80	99.0	0.80
19.98	**99.9**	**0.86**
20.00	**100.0**	**1.06**
20.02	**100.1**	**1.26**
20.20	101.0	1.32
40.00	200.0	1.44

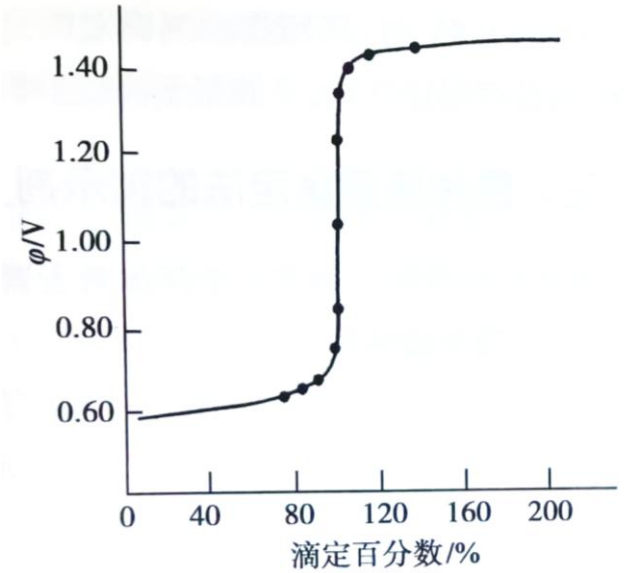


图 5-1 Ce^{4+}溶液滴定 Fe^{2+}溶液的滴定曲线

Section 2 Basic Principles of Redox Titration

2 Redox titration curve氧化还原滴定曲线

Reaction: $Ce^{4+} + Fe^{2+} \rightleftharpoons Ce^{3+} + Fe^{3+}$

When an oxidant is used to titrate a reductant, the potential at a titration percentage of 50% is the conditional potential of the reductant couple, and the potential at a titration percentage of 200% is the conditional potential of the oxidant couple. Near the stoichiometric point (±0.1%), the electrode potential shows a distinct jump (0.86~1.26 V), and the stoichiometric point potential φsp lies exactly in the middle of the titration jump.

用氧化剂滴定还原剂时，滴定百分数为 50% 时的电位是还原剂电对的条件电位，滴定百分数为 200% 时的电位是氧化剂电对的条件电位；在化学计量点附近 (前后 0.1%)，电极电位有明显的突跃 (0.86~1.26V)，化学计量点电位φ_{sp}恰好在滴定突跃的中间

Section 2 Basic Principles of Redox Titration

2 Redox titration curve氧化还原滴定曲线

Reaction: $Ce^{4+} + Fe^{2+} \rightleftharpoons Ce^{3+} + Fe^{3+}$

General formula for symmetric redox couples对称电对:

$$\varphi_{sp} = \frac{n_1\varphi_1^{\ominus\prime} + n_2\varphi_2^{\ominus\prime}}{n_1 + n_2}$$

If the numbers of transferred electrons of the two redox couples are unequal, φ_{sp} is not at the midpoint of the potential jump and leans toward the couple with a larger number of transferred electrons.

若两电对电子转移数不等，φ_{sp}不在突跃正中，偏向电子转移数多的一方。

Section 2 Basic Principles of Redox Titration

2 Redox titration curve氧化还原滴定曲线

Reaction: $Ce^{4+} + Fe^{2+} \rightleftharpoons Ce^{3+} + Fe^{3+}$

Formula for the jump range:

$$\varphi_2^{\ominus\prime} + \frac{0.059 \times 3}{n_2} \sim \varphi_1^{\ominus\prime} - \frac{0.059 \times 3}{n_1}$$

The larger the conditional potential difference $\Delta\varphi^{\circ\prime}$ → the wider the jump range → the easier it is to select an indicator and the more accurate the measurement result.

条件电位差 $\Delta\varphi^{\circ\prime}$ 越大 → 突跃范围越宽 → 选择指示剂越容易、测定结果越准确

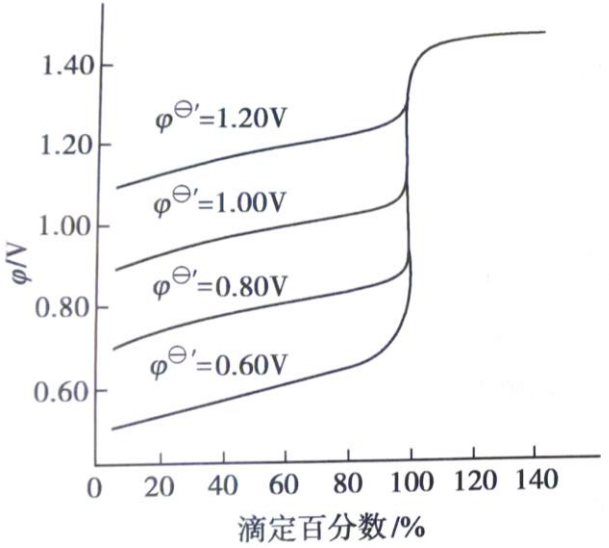


图 5-2 Ce^{4+}溶液滴定 4 种还原剂的滴定曲线

Section 2 Basic Principles of Redox Titration

2 Redox titration curve氧化还原滴定曲线

$\Delta\varphi^{\circ\prime} > 0.3\sim0.4$ V

The jump is obvious, and redox indicators can be used for visual endpoint detection.

0.2 ~ 0.3 V

The jump is relatively small, and potentiometry is recommended for endpoint determination.

$\Delta\varphi^{\circ\prime} < 0.2$ V

No obvious jump is observed, so it cannot be applied to titrimetric analysis.

Section 2 Basic Principles of Redox Titration

3 Indicators for redox titration氧化还原滴定法的指示剂

(1) Self indicator自身指示剂

some standard solutions or analytes are deeply coloured while products are colourless or pale; no additional indicator is needed and their own colour change indicates endpoint

$MnO_4^- \rightarrow Mn^{2+}$

purple **colourless**

ep： pink

$I_3^- \rightarrow I^-$

dark **colourless**

ep： pale yellow

advantage: no additional indicator is needed; its own colour change indicates endpoint

优点：无需选择指示剂，利用自身颜色变化指示终点

Section 2 Basic Principles of Redox Titration

3 Indicators for redox titration

(2) Special indicator特殊指示剂

some substances are not redox-active but react with oxidants or reductants to produce a special reversible colour change indicating endpoint

Example: starch淀粉 + I_3^-—— deep blue complex

characteristics: sensitive and reversible; used in direct or indirect iodimetry

特点：反应灵敏、可逆，应用于直接或间接碘量法

Section 2 Basic Principles of Redox Titration

3 Indicators for redox titration

(3) Redox indicator氧化还原指示剂

本身是弱氧化剂或弱还原剂，其氧化态（In_{Ox}）和还原态（In_{Red}）具有不同的颜色，在滴定过程中因被氧化或还原而发生颜色变化来指示终点

It is a weak oxidizing agent or weak reducing agent. Its oxidized form (In_{Ox}) and reduced form (In_{Red}) have different colors, and color changes occur during titration due to oxidation or reduction to indicate the endpoint.

$$In_{Ox} + ne \rightleftharpoons In_{Red}$$

$$\varphi_{In_{Ox}/In_{Red}} = \varphi^{\ominus'}_{In_{Ox}/In_{Red}} + \frac{0.059}{n} \lg \frac{c_{In_{Ox}}}{c_{In_{Red}}} \qquad (25℃)$$

$\varphi^{\ominus'}_{In_{Ox}/In_{Red}}$ *is* conditional potential of the indicator

Section 2 Basic Principles of Redox Titration

3 Indicators for redox titration

(4) External indicator外指示剂

- Some substances are inherently redox-active and can undergo redox reactions with the standard solution or the analyte solution.
- For this reason, they cannot be added into the analyte solution. Instead, near the stoichiometric point, a glass rod is used to take a drop of the titrated solution to react with the substance externally.
- The titration endpoint is determined based on colour changes. Such substances are known as outside indicators.

有的物质本身具有氧化还原性，能与标准溶液或被测溶液发生氧化还原反应，故不能加入被测溶液中，只能在化学计量点附近，用玻棒蘸取被滴定的溶液在外面与其作用，根据颜色变化来判断滴定终点，这类物质称外指示剂

Section 2 Basic Principles of Redox Titration

3 Indicators for redox titration

(5) Irreversible indicator不可逆指示剂

- Some substances react with a slight excess of standard solution to produce an irreversible color change, which is used to indicate the titration endpoint.
- Such substances are called irreversible indicators.
- For example, in the bromate method, excess potassium bromate solution releases bromine under acidic conditions. Bromine destroys the chromogenic structure of methyl red or methyl orange indicators, resulting in a color change to indicate the titration endpoint.

有的物质与微过量标准溶液作用，可发生不可逆的颜色变化，从而指示滴定终点，这类物质称为不可逆指示剂。例如，溴酸钾法是利用过量的溴酸钾溶液在酸性条件中析出溴，溴能破坏甲基红或甲基橙指示剂的显色结构而产生颜色变化来指示滴定终点

Section 2 Basic Principles of Redox Titration

4 Endpoint error of redox titration

the error caused by the difference between endpoint滴定终点(ep) and stoichiometric point化学计量点(sp) is called endpoint error or titration error滴定终点误差或滴定误差

Ringbom error equation:

$$Ox_1 + Red_2 \rightleftharpoons Ox_1 + Red_2$$

$$TE(\%) = \frac{[Ox_1]_{ep} - [Red_2]_{ep}}{c_{sp}} \times 100\%$$

$$TE(\%) = \frac{10^{\Delta\varphi/0.059} - 10^{-\Delta\varphi/0.059}}{10^{\Delta\varphi^{\ominus\prime}/2\times0.059}} \times 100\%$$

$\varphi^{\ominus\prime}$the larger, TE(%) the smaller; $\Delta\varphi$ the smaller,TE(%) the smaller

Section 2 Basic Principles of Redox Titration

4 Endpoint error of redox titration

if the electron-transfer numbers of the two half-reactions are unequal ($n_1 \neq n_2$), but when both couples are still symmetric, endpoint error equation is:

$$TE(\%) = \frac{10^{n_1 \cdot \Delta\varphi / 0.059} - 10^{-n_2 \cdot \Delta\varphi / 0.059}}{10^{n_1 \cdot n_2 \cdot \Delta\varphi^{\ominus\prime} / (n_1 + n_2) \times 0.059}} \times 100\%$$

Section 2 Basic Principles of Redox Titration

4 Endpoint error of redox titration

Example5-5: Titrate 0.1000 mol/L Fe^{2+} sample solution with 0.1000 mol/L Ce^{4+} standard solution in 1.0 mol/L H_2SO_4 solution. Calculate the titration end point error if sodium diphenylaminesulfonate二苯胺磺酸钠 is selected as the indicator.
(Given： $\varphi^{\ominus\prime}_{Ce^{4+}/Ce^{3+}}$=1.44V； $\varphi^{\ominus\prime}_{Fe^{3+}/Fe^{2+}}$=0.68V； conditional potential of sodium diphenylamine sulfonate $\varphi^{\ominus\prime}_{In}$ =0.84V）

Solution: the titration reaction; number of electrons transferred $n_1 = n_2 = 1$

$$Ce^{4+} + Fe^{2+} \rightleftharpoons Ce^{3+} + Fe^{3+}$$

$$\triangle\varphi^{\ominus\prime} = 1.44 - 0.68 = 0.76V$$

$$\varphi_{sp} = \frac{n_1\varphi_1^{\ominus\prime} + n_2\varphi_2^{\ominus\prime}}{n_1 + n_2} = \frac{1.44 + 0.68}{2} = 1.06V$$

Section 2 Basic Principles of Redox Titration

4 Endpoint error of redox titration

$$\varphi_{ep} = 0.84\text{V}$$

$$\Delta\varphi = \varphi_{ep} - \varphi_{sp} = 0.84 - 1.06 = -0.22\text{V}$$

$$TE(\%) = \frac{10^{\Delta\varphi/0.059} - 10^{-\Delta\varphi/0.059}}{10^{\Delta\varphi^{\ominus'}/2\times0.059}} \times 100\% = \frac{10^{-0.22/0.059} - 10^{0.22/0.059}}{10^{0.76/2\times0.059}} \times 100\%$$

$$= \text{-0.2\%}$$

Section 2 Basic Principles of Redox Titration

5 Pretreatment of redox titration

可被滴定价态

the process of converting the analyte into a titratable oxidation state

the oxidant or reductant used in pretreatment must meet the following requirements:

(1) can rapidly, quantitatively and completely oxidise or reduce the analyte to the specified oxidation state

(2) the reaction has selectivity and does not form interfering substances

(3) excess oxidant or reductant should be easy to remove

（1）能将待测组分迅速、定量、完全地氧化或还原为指定的价态
（2）反应具有一定的选择性，不生成干扰测定的物质
（3）过量的氧化剂过还原剂应易于除去

Section 3 Common Redox Titration Methods

Section 3 Common Redox Titration Methods

1 Iodimetry碘量法

(1) Iodimetrybasic principles碘量法的基本原理

A redox titration method utilizing the oxidizing property of I_2 or the reducing property of I^-

redox couple reaction I_2（s）+ 2e $\rightleftharpoons$ 2I- $\varphi^{\ominus}_{I_2/I^-} = 0.535V$

$$I_2 + I^- \rightleftharpoons I_3^-$$

$$I_3^- + 2e \rightleftharpoons 3I^- \qquad \varphi^{\ominus}_{I_3^-/I^-} = 0.545V$$

$I_2$is a relatively weak oxidant, $I^-$is a moderate reductant

I_2是较弱的氧化剂，I^-是中等强度的还原剂

Section 3 Common Redox Titration Methods

1 Iodimetry

(1) Basic principles of iodimetry

1. Titration mode

(1) direct iodimetry (iodimetric titration):直接碘量法（碘滴定法）

I_2 weak oxidant 弱氧化剂

analytes: reducing substances

Can determine:S^{2-}、Sn（II）、SO_3^{2-}、As (III), vitamin C etc.

测定物：具有还原性物质

可测：S^{2-}、Sn（II）、SO_3^{2-}、As（III）、维生素C等

Section 3 Common Redox Titration Methods

1 Iodimetry

(1) Basic principles of iodimetry

$I_2 + 2S_2O_3^{2-} \rightleftharpoons 2I^- + S_4O_6^{2-}$(weakly acidic)

1. titration mode

(2) indirect iodimetry间接碘量法（滴定碘法）: using I^- moderate reducing power to titrate oxidising substances利用I^-的中等强度还原性滴定氧化性物质

①displacement iodimetry置换碘量法: I^- oxidised to I_2 by the oxidising substance , using $Na_2S_2O_3$ standard solution to titrate the generated I_2；②residual iodimetry剩余碘量法: reducing substance reacts with a measured excess of I_2 standard solution, then use $Na_2S_2O_3$ standard solution to titrate the residual I_2

Section 3 Common Redox Titration Methods

1 Iodimetry

(1) Basic principles of iodimetry

1.titration mode

displacement iodimetry of Analytes置换碘量法的测定物: MnO_4^-、$Cr_2O_7^{2-}$、BrO_3^-、IO_3^-、H_2O_2、Cl_2、Br_2、Cu^{2+}, bleaching powder, chloramine T and other oxidising substances漂白粉、氯胺T等氧化性物质

residual iodimetry剩余碘量法analytes: methionine, sodium metabisulfite, glucose, formaldehyde, thiourea and other reducing substances

甲硫氨酸、焦亚硫酸钠、葡萄糖、甲醛及硫脲

Section 3 Common Redox Titration Methods

1 Iodimetry

(1) Basic principles of iodimetry

1.titration mode

Example: Determine copper salt by indirect iodimetry: accurately weigh blue vitriol胆矾 sample (containing $CuSO_4 \cdot 5H_2O$） 0.5580g, place in conical flask, dissolve in water, acidification by HAc, add excess KI, the liberated I_2 was titrated by 0.1020mol/L $Na_2S_2O_3$ standard solution; at endpoint, 20.58ml of $Na_2S_2O_3$ standard solution was consumed; Find the mass fraction of $CuSO_4 \cdot 5H_2O$ in the sample ($M_{CuSO_4 \cdot 5H_2O}$ = 249.68g/mol）

Section 3 Common Redox Titration Methods

1 Iodimetry

1.titration mode

Solution: the reaction is: $2Cu^{2+} + 4I^- \rightleftharpoons 2CuI\downarrow + I_2$

$$I_2 + 2S_2O_3^{2-} \rightleftharpoons 2I^- + S_4O_6^{2-}$$

therefore 2mol Cu^{2+} ⌒ 1mol I_2 ⌒ 2 $Na_2S_2O_3^{2-}$

$$w_{CuSO_4}(\%) = \frac{(cV)_{Na_2S_2O_3} \times M_{CuSO_4}}{m_s \times 1000} \times 100\%$$

$$= \frac{0.1020 \times 20.58 \times 249.68}{0.5580 \times 1000} \times 100\% = 93.9\%$$

Section 3 Common Redox Titration Methods

1 Iodimetry

(1) Basic principles of iodimetry

2.titration conditions

(1) titration conditions for direct iodimetry:

this method can only be performed inperform in acidic, neutral or weakly alkaline solution if the solution pH＞9, the following disproportionation reaction occurs:

$3I_2 + 6OH^- \rightleftharpoons 5I^- + IO_3^- + 3H_2O$ (disproportionation reaction歧化反应）

直接碘量法的滴定条件：该法只能在酸性，中性或弱碱性溶液中进行

如果溶液的pH＞9，则会发生歧化反应：

Section 3 Common Redox Titration Methods

1 Iodimetry

2.titration conditions

（2）titration conditions for indirect iodimetry: $Na_2S_2O_3$ titration I_2 reaction is generally performed in neutral or weakly acidic solution

alkaline solution: I^- and $S_2O_3^{2-}$ side reactions

$$S_2O_3^{2-} + 4I_2 + 10OH^- \rightleftharpoons 2SO_4^{2-} + 8I^- + 5H_2O$$

$$3I_2 + 6OH^- \rightleftharpoons IO_3^- + 5I^- + 3H_2O$$

strongly acidic solution: $S_2O_3^{2-}$ decomposes, I^- undergoes oxidation→causes delayed endpoint

$$S_2O_3^{2-} + 2H^+ \rightleftharpoons SO_2\uparrow + S\downarrow + H_2O$$

$$4I^- + O_2 + 4H^+ \rightleftharpoons 2I_2 + 2H_2O$$

Section 3 Common Redox Titration Methods

1 Iodimetry

(1) Basic principles of iodimetry

①加入过量的KI，使I_2生成I_3^-，增大I_2溶解度，减少I_2的挥发；②滴定在室温下进行，温度升高会加快I_2的挥发；③析出I_2的反应可在碘量瓶中进行，快滴慢摇

3.error control误差控制

(1) Methods to prevent the volatilization of I_2防止I_2挥发的方法:

① Add excess KI to allow I_2 to form I_3^-, increasing the solubility of I_2 and reducing its volatilization; ② Perform the titration at room temperature, as elevated temperature accelerates the volatilization of I_2; ③ Carry out the reaction that liberates I_2 in an iodine flask, with rapid titration and slow shaking.

Section 3 Common Redox Titration Methods

1 Iodimetry

(1) Basic principles of iodimetry

3.error control

防止I^-被空气中的O_2氧化的方法

①控制溶液的酸度，因为酸度增大会加快O_2氧化I^-；②除去对I^-的氧化起催化作用的物质；③光照会加速O_2氧化I^-，故密塞避光放置，析出I_2的反应在完全后立即滴定，快滴慢摇

(2) Methods to prevent I^- from being oxidized by O_2 in the air: ① Control the acidity of the solution, because increased acidity accelerates the oxidation of I^- by O_2; ② Remove substances such as Cu^{2+} and NO_2^- that catalyze the oxidation of I^-; ③ Light accelerates the oxidation of I^- by O_2, so the solution should be tightly stoppered and stored away from light. Perform titration immediately after the reaction that generates I_2 is completed, with rapid titration and slow shaking.

Section 3 Common Redox Titration Methods

1 Iodimetry

(1) Basic principles of iodimetry

4. Indicator

(1) self-indicator

I_2 standard solution——direct iodimetry

(2) starch indicator

Requirements: freshly prepared, room temperature, weakly acidic solution

Colour-change principle: $I_2 + I^- \rightleftharpoons I_3^-$ (sensitive and reversible)

(forms deep-blue complex with starch)

与淀粉形成深蓝色配合物

Section 3 Common Redox Titration Methods

1 Iodimetry

(1) Basic principles of iodimetry

4.indicator

(2) starch indicator

注：间接碘量法中若指示剂加入过早，则溶液中大量的I_2被淀粉表面牢固地吸附，使蓝色褪去缓慢（称终点迟钝）而产生误差

Addition time: direct iodimetry——add before titration (endpoint: blue forms)

indirect iodimetry——add near endpoint (endpoint: blue disappears)

Note: In the indirect iodometric method, if the indicator is added too early, a large amount of I_2 in the solution will be firmly adsorbed on the surface of starch, resulting in slow fading of the blue color (known as endpoint sluggishness) and thus causing errors.

Section 3 Common Redox Titration Methods

1 Iodimetry

(2) Applications of iodimetry

1.preparation and standardisation of solution

(1) Preparation and standardization of iodine standard solution:

I_2 is volatile and corrosive, so iodine standard solution is usually prepared by standardization method; add concentrated KI solution; add a small amount of hydrochloric acid; the I_2 solution shall be stored in a brown bottle with a tightly sealed stopper.

I_2具有挥发性和腐蚀性，通常采用标定法配制碘标准溶液；加入KI的浓溶液；加入少量盐酸；I_2溶液应贮于棕色瓶中，密塞存放

Section 3 Common Redox Titration Methods

(2) preparation and standardisation of sodium thiosulfate硫代硫酸钠standard solution

- indirect method
- $Na_2S_2O_3$ is unstable and decomposes because:

  a. function of dissolved CO_2 in water:$S_2O_3^{2-} + CO_2 + H_2O \rightarrow HSO_3^- + HCO_3^- + S\downarrow$

  嗜硫细菌

  b. effect of microorganisms such as sulfur-oxidizing bacteria present in water:$S_2O_3^{2-} \rightarrow SO_3^{2-} + S\downarrow$

  c. function of dissolved O_2 in water: $2S_2O_3^{2-} + O_2 \rightarrow 2SO_4^{2-} + 2S\downarrow$

- preparation: freshly boiled cooled distilled water; add a littleNa_2CO_3makes pH9～10, store solution in brown bottle for a period[7～10days,《Chinese Pharmacopoeia》（2020edition) specifies standing for1months] , standardise

Section 3 Common Redox Titration Methods

1 Iodimetry

(2) Applications of iodimetry

1.preparation and standardisation of solution

(2) preparation and standardisation of sodium thiosulfate硫代硫酸钠standard solution

- standardisation: primary standard基准物质 ($K_2Cr_2O_7$、$KBrO_3$、KIO_3etc.)

$$Cr_2O_7^{2-} + 6I^- + 14H^+ \rightleftharpoons 2Cr^{3+} + 3I_2 + 7H_2O$$ (displacement reaction置换反应)

$$I_2 + 2S_2O_3^{2-} \rightleftharpoons 2I^- + S_4O_6^{2-}$$ (titration reaction滴定反应)

Section 3 Common Redox Titration Methods

1 Iodimetry

Example Standardize the concentration of the $Na_2S_2O_3$ standard solution using $K_2Cr_2O_7$ primary standard: Weigh out 0.5012 g of $K_2Cr_2O_7$ primary standard, dissolve it in water and dilute to 100.0 mL. Pipette 20.00 mL of this solution, add H_2SO_4 and KI solutions, then titrate with the $Na_2S_2O_3$ solution to be standardized to the endpoint. The volume consumed is 20.05 mL. Calculate the concentration of the $Na_2S_2O_3$ standard solution. ($M_{K_2Cr_2O_7} = 294.18 g/mol$)

$$Cr_2O_7^{2-} + 6I^- + 14H^+ \rightleftharpoons 2Cr^{3+} + 3I_2 + 7H_2O$$

Solution:

$$I_2 + 2S_2O_3^{2-} \rightleftharpoons 2I^- + S_4O_6^{2-}$$

1mol $Cr_2O_7^{2-}$ ≏ 3mol I_2 ≏ 6mol $S_2O_3^{2-}$, therefore $n_{S_2O_3^{2-}} = 6n_{Cr_2O_7^{2-}}$

$$c_{Na_2S_2O_3} = \frac{6 \times \frac{m_{K_2Cr_2O_7}}{M_{K_2Cr_2O_7}} \times \frac{20.00}{100.0}}{V_{Na_2S_2O_3} \times 10^{-3}} = \frac{6 \times \frac{0.5012}{294.18} \times \frac{20.00}{100.0}}{20.05 \times 10^{-3}} = 0.1020 mol/L$$

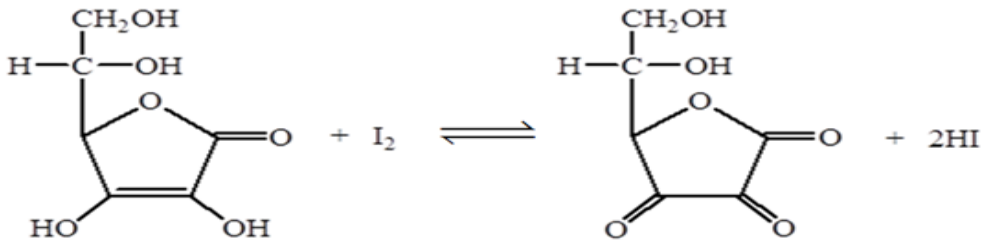
Section 3 Common Redox Titration Methods

1 Iodimetry

(2) Applications of iodimetry

2. application example

Example: Determination of Vitamin C Content (Direct Iodimetry)



Note: Titration is usually carried out in an acetic acid medium, protected from light and heat. 通常在醋酸介质中进行滴定，避光、避热

Section 3 Common Redox Titration Methods

2 Permanganate titration高锰酸钾法

(1) Permanganate method basic principles

redox titration using potassium permanganate as titrant

$$MnO_4^- + 5e + 8H^+ \rightleftharpoons Mn^{2+} + 4H_2O \qquad \varphi^{\ominus} = 1.51V$$

Note: adjust acidity with H_2SO_4 , do not use HCl or HNO_3

Section 3 Common Redox Titration Methods

2 Permanganate titration

(1) Basic principles of permanganate titration

1.direct titration

such as Fe^{2+} 、$C_2O_4^{2-}$、As（III）、H_2O_2etc.

2.residual titration mode

such as MnO_2、PbO_2 、CrO_4^{2-}、 BrO_3^-、IO_3^-

3.indirect titration

determine some non-redox substances such as Ca^{2+}

Section 3 Common Redox Titration Methods

2 Permanganate titration

(2) Applications of permanganate titration

(1) preparation and standardisation of standard solution

(2) standardisation$KMnO_4$primary standards for the solution:As_2O_3、$H_2C_2O_4 \cdot 2H_2O$、$Na_2C_2O_4$etc.

$$2MnO_4^- + 5\ C_2O_4^{2-} + 16H^+ \rightleftharpoons 2Mn^{2+} + 10CO_2\uparrow + 8H_2O$$

Notes during standardisation: (1) temperature:75～85℃ ; (2) acidity; (3) titration rate

标定时应注意下列条件：①温度：75～85℃；②酸度；③滴定速度

Section 3 Common Redox Titration Methods

2 Permanganate titration

(2) Applications of permanganate titration

Example Determination of hydrogen peroxide content

$$2MnO_4^- + 5H_2O_2 + 16H^+ \rightleftharpoons 2Mn^{2+} + 5O_2\uparrow + 8H_2O$$

- Dilute hydrogen peroxide appropriately
- Perform titration at room temperature
- Add titrant slowly at first, then faster
- A small amount of Mn^{2+} can be added as a catalyst

- 适当稀释过氧化氢
- 室温进行滴定
- 滴定先慢后快
- 可加入少量Mn^{2+}作催化剂

Section 3 Common Redox Titration Methods

2 Permanganate titration

(2) Applications of permanganate titration

Example Determination of Ferrous Sulfate Content

$$5Fe^{2+} + MnO_4^- + 8H^+ \rightleftharpoons 5Fe^{3+} + Mn^{2+} + 4H_2O$$

- Fe^{2+} is easily oxidized by O_2 in the air, so titration should be carried out immediately after sample dissolution.
- $KMnO_4$ acts as a self-indicator (ferroin can also be used).
- It is only applicable to the assay of ferrous sulfate raw material drug.
- Cerimetry can be adopted for the determination of ferrous sulfate preparations.

- Fe^{2+}易被空气中的O_2氧化，样品溶解后应立即滴定
- $KMnO_4$作自身指示剂（也可用邻二氮菲亚铁）
- 只适用于测定硫酸亚铁原料药
- 测定硫酸亚铁制剂可改用铈量法

Section 3 Common Redox Titration Methods

3 Sodium nitrite titration

(1) Sodium nitrite method basic principles

It is a redox titration method that uses sodium sulfite as the titrant, relying on the diazotization reaction or nitrosation reaction between sodium sulfite and organic amines.

是以亚硫酸钠为滴定剂，利用亚硫酸钠与有机胺类物质发生重氮化反应或亚硝基化反应的氧化还原滴定法

Section 3 Common Redox Titration Methods

3 Sodium nitrite titration

(1) Basic principles of sodium nitrite titration

1.diazotisation titration

$$ArNH_2 + NaNO_2 + 2HCl \rightleftharpoons [Ar—N^+\equiv N]Cl^- + NaCl + 2H_2O$$

（1）type and concentration of acid: reaction rate：

HBr＞HCl＞H_2SO_4/HNO_3, hydrochloric acid is commonly used

Acidity: too high affects rate; too low gives low result

Section 3 Common Redox Titration Methods

3 Sodium nitrite titration

(1) Basic principles of sodium nitrite titration

（2）titration rate and temperature

$$3HNO_2 \rightarrow HNO_3 + H_2O + 2NO\uparrow$$

$$[Ar - N^+ \equiv N]Cl^- + H_2O \rightarrow Ar - OH + N_2\uparrow + HCl$$

- 一般规定滴定在15℃以下进行。《中华人民共和国药典》（2020年版）规定可在室温（10～30℃）下采用“快速滴定法”进行
- 液面下生成的HNO_2迅速扩散并立即与芳伯胺作用，可有效防止HNO_2的逸失和分解，提高测定结果的准确度

- Titration is generally specified to be carried out below 15℃. The Pharmacopoeia of the People‘s Republic of China (2020 Edition) stipulates that the “rapid titration method” can be adopted at room temperature (10～30℃).
- The HNO_2 generated below the liquid surface diffuses rapidly and immediately reacts with aromatic primary amines, which can effectively prevent the escape and decomposition of HNO_2 and improve the accuracy of measurement results.

Section 3 Common Redox Titration Methods

3 Sodium nitrite titration

(1) Basic principles of sodium nitrite titration

（3）substituents on the benzene ring

- electron-withdrawing groups accelerate reaction:

such as—NO_2、— SO_3H、—COOH、—X etc.

- electron-donating group slow downreaction rate:

such as—CH_3、— OH、—OR etc.

- an appropriate amount KBr can catalyse the reaction

Section 3 Common Redox Titration Methods

3 Sodium nitrite titration

(1) Basic principles of sodium nitrite titration

2. Nitrosation titration

$$ArNHR + NaNO_2 + HCl \rightleftharpoons ArN（NO）R + NaCl + H_2O$$

Application:

- diazotisation titration: aromatic primary amine drugs,such as sulfonamides, procaine hydrochloride, benzocaine, dapsone
- nitrosation titration: aromatic secondary amine drugs such as primaquine phosphate and tetracaine hydrochloride

- 重氮化滴定法：芳伯胺类药物，如磺胺类药物、盐酸普鲁卡因、苯佐卡因、氨苯砜等
- 亚硝基化滴定法：芳仲胺类药物，如磷酸伯氨喹、盐酸丁卡因等

Section 3 Common Redox Titration Methods

3 Sodium nitrite titration

(1) Basic principles of sodium nitrite titration

3.indicator

(1) external indicator

potassium iodide containing zinc chloride——starch paste or test paper

含氯化锌的碘化钾——淀粉糊或试纸

$$2NO_2^- + 2I^- + 4H^+ \rightleftharpoons I_2 + 2NO\uparrow + 2H_2O$$

(2) internal indicator

Widely used ones include Orange IV-methylene blue neutral red, brilliant cresyl blue and diphenylamine应用较多的有橙黄IV-亚甲蓝中性红、亮甲酚蓝及二苯胺

Section 3 Common Redox Titration Methods

3 Sodium nitrite titration

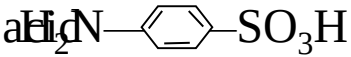
(2)applications of sodium nitrite titration

1.preparation of sodium nitrite standard solution

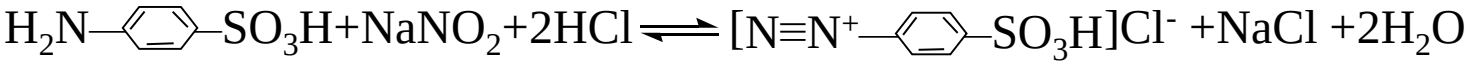
Preparation: indirect method

store in brown bottle, tightly closed, protected from light

2.standardisation of sodium nitrite standard solution

primary standard: sulfanilic acid 

Dead-stop titration indicates endpoint:



Section 3 Common Redox Titration Methods

4 Potassium bromate and bromine methods溴酸钾法和溴量法

(1) Potassium bromate method

$$BrO_3^- + 6H^+ + 6e \rightleftharpoons Br^- + 3H_2O \qquad \varphi^{\ominus}_{BrO_3^-/Br^-} = 1.44V$$

- Determination of reducing substances in acidic solutions
- Add the indicator near the endpoint: methyl red / methyl orange (structure destruction — fading of red colour)
- Bromate method can directly determine reducing substances such as ferrous salts, cuprous salts, arsenites, stannites, iodides and imines.

- 在酸性溶液中测定还原性物质
- 临近终点时加入指示剂：甲基红/甲基橙（破坏结构——红色褪去）
- 溴酸钾法可直接测定亚铁盐、亚铜盐、亚砷酸盐、亚锡盐、碘化物和亚胺类等还原性物质

Section 3 Common Redox Titration Methods

4 Potassium bromate and bromine methods

(2) Bromine method

Bromine solution: $KBrO_3$ and KBr generate a quantitative amount of Br_2

$$BrO_3^- + 5Br^- + 6H^+ \rightleftharpoons 3Br_2 + 3H_2O$$

$$Br_2 + 2I^- \rightleftharpoons I_2 + 2Br^-$$

- bromine method can determine SO_2、H_2S, sulfites and hydroxylamine
- using8-hydroxyquinoline as precipitant, bromine method can also indirectly determine metal ions such as Al^{3+}、Mg^{2+} and Fe^{3+}

Section 3 Common Redox Titration Methods

5 Potassium dichromate titration重铬酸钾法

potassium dichromate methodbasic principles

$$Cr_2O_7^{2-} + 6e + 14H^+ \rightleftharpoons 2Cr^{3+} + 7H_2O$$

Medium: HCl (not affected byCl^-reducing property)

Characteristics:$K_2Cr_2O_7$ solid reagent is easy to purify and stable—prepare directly $K_2Cr_2O_7$ standard solution

Application: determination ofFe^{2+}

$$Cr_2O_7^{2-} + 6Fe^{2+} + 14H^+ \rightleftharpoons 2Cr^{3+} + 6Fe^{3+} + 7H_2O$$

$$Cr_2O_7^{2-} \backsimeq 6Fe^{2+}$$

Section 3 Common Redox Titration Methods

6 Cerimetry铈量法

It is a redox titration method using Ce^{4+} solution as the standard solution. The half-cell reaction is

是以Ce^{4+}溶液为标准溶液的氧化还原滴定法。半电池反应为：

$$Ce^{4+} + e \rightleftharpoons Ce^{3+} \qquad \varphi^{\ominus}_{Ce^{4+}/Ce^{3+}} = 1.45V$$

Compared with the $KMnO_4$ method, the advantages of the cerium sulfate titration method:
(1) Standard cerium sulfate solution can be prepared directly and has good stability
(2) High selectivity
(3) Simple reaction mechanism with few side reactions

与$KMnO_4$法比较，铈量法的优点：

（1）硫酸铈标准溶液可直接配制，稳定性好

（2）选择性高

（3）反应机制简单，副反应少

Section 3 Common Redox Titration Methods

7 Potassium periodate method高碘酸钾法

$$IO_6^{5-} + H^+ + 2e \rightleftharpoons IO_3^- + 3H_2O \qquad \varphi^{\ominus}_{H_5IO_6/IO_3} = 1.60V$$

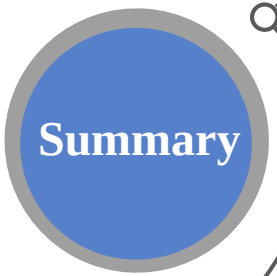
Standardisation: $IO_4^- + 7I^- + 8H^+ \rightleftharpoons 4I_2 + 4H_2O$

$$IO_3^- + 5I^- + 6H^+ \rightleftharpoons 3I_2 + 3H_2O$$

The potassium periodate method can be used for the determination of organic compounds such as α-hydroxy alcohols, α-amino alcohols, α-carbonyl alcohols and polyhydric alcohols (e.g., glycerol, mannitol) in acidic media.

高碘酸钾法在酸性介质中可用于α-羟基醇、α-氨基醇、α-羰基醇、多羟基醇（如甘油、甘露醇）等有机物的测定

Chapter summary



1.redox equilibrium

conditional potential, influencing factors and calculation; conditional equilibrium constant

2.indicators for redox titration

classification, endpoint indication principles and selection of redox indicators

3.redox titration curve

factors affecting potential jump range; calculation of stoichiometric point and jump range

4.iodimetry, permanganometry, sodium nitrite titration

principles, titration conditions, indicators, endpoint, standard solutions, primary standards, calculations

本章小结

小结

- **1.氧化还原平衡**

  条件电位、影响因素及计算；条件平衡常数及计算

- **2.氧化还原滴定法的指示剂**

  氧化还原指示剂的分类、指示终点的原理和选择原则

- **3.氧化还原滴定曲线**

  影响电位突跃范围的因素；化学计量点和突跃范围的计算

- **4.碘量法、高锰酸钾法、亚硝酸钠法**

  基本原理、滴定条件、指示剂、终点、标准溶液、基准物、氧化还原滴定结果的计算

Thank you for watching